Java and JVM

Practical task

1. Download/install Tomcat server.
2. Verify that it works by visiting the root page.
   What ports are used by the java process?
3. Remove all default applications (including manager), restart Tomcat.
4. Download Jenkins WAR and deploy into Tomcat.
5. Verify that application works (visit application URL).
6. Enable JMX in Tomcat.
   What ports are used by the java process?
   Change CATALINA_OPTS to use same for RMI as for JMX
   What ports are used by the java process?
7. Rerun tomcat with min heap size 10M and max heap size 20M.
   What type of error will you get?
   Increase min heap size to 1G and max heap size to 3G, enable parallel garbage collector.
8. Connect by JConsole to Tomcat and look around.
9. Stop Tomcat.
10. Launch Jenkins WAR as a standalone application, verify that it works.

Tomcat + Jenkins Practical Task

Objective

This demonstrates installation and configuration of Apache Tomcat, deployment of Jenkins, JVM tuning, JMX monitoring, and standalone execution.

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1. Install Java (Prerequisite)

Tomcat requires Java to run.

```
- brew install openjdk@17
```

Set environment variables:

```
- export JAVA_HOME=$(/usr/libexec/java_home)
- export PATH=$JAVA_HOME/bin:$PATH
```

Verify:

```
- java -version
```

Output:

```
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment Homebrew (build 17.0.18+0)
OpenJDK 64-Bit Server VM Homebrew (build 17.0.18+0, mixed mode,
sharing)
```

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2. Install and Start Tomcat

```
- tar -xvf apache-tomcat-9.0.117.tar.gz
- cd apache-tomcat-9.0.117/bin
- chmod +x *.sh

  Start Tomcat :
    - ./startup.sh
```

Output:

```
Using CATALINA_BASE:   /Users/npahwa/Downloads/apache-tomcat-9.0.117
Using CATALINA_HOME:   /Users/npahwa/Downloads/apache-tomcat-9.0.117
Using CATALINA_TMPDIR:
/Users/npahwa/Downloads/apache-tomcat-9.0.117/temp
Using JRE_HOME:
/opt/homebrew/Cellar/openjdk@17/17.0.18/libexec/openjdk.jdk/Contents/H
ome
Using CLASSPATH:
/Users/npahwa/Downloads/apache-tomcat-9.0.117/bin/bootstrap.jar:/Users
/npahwa/Downloads/apache-tomcat-9.0.117/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
```

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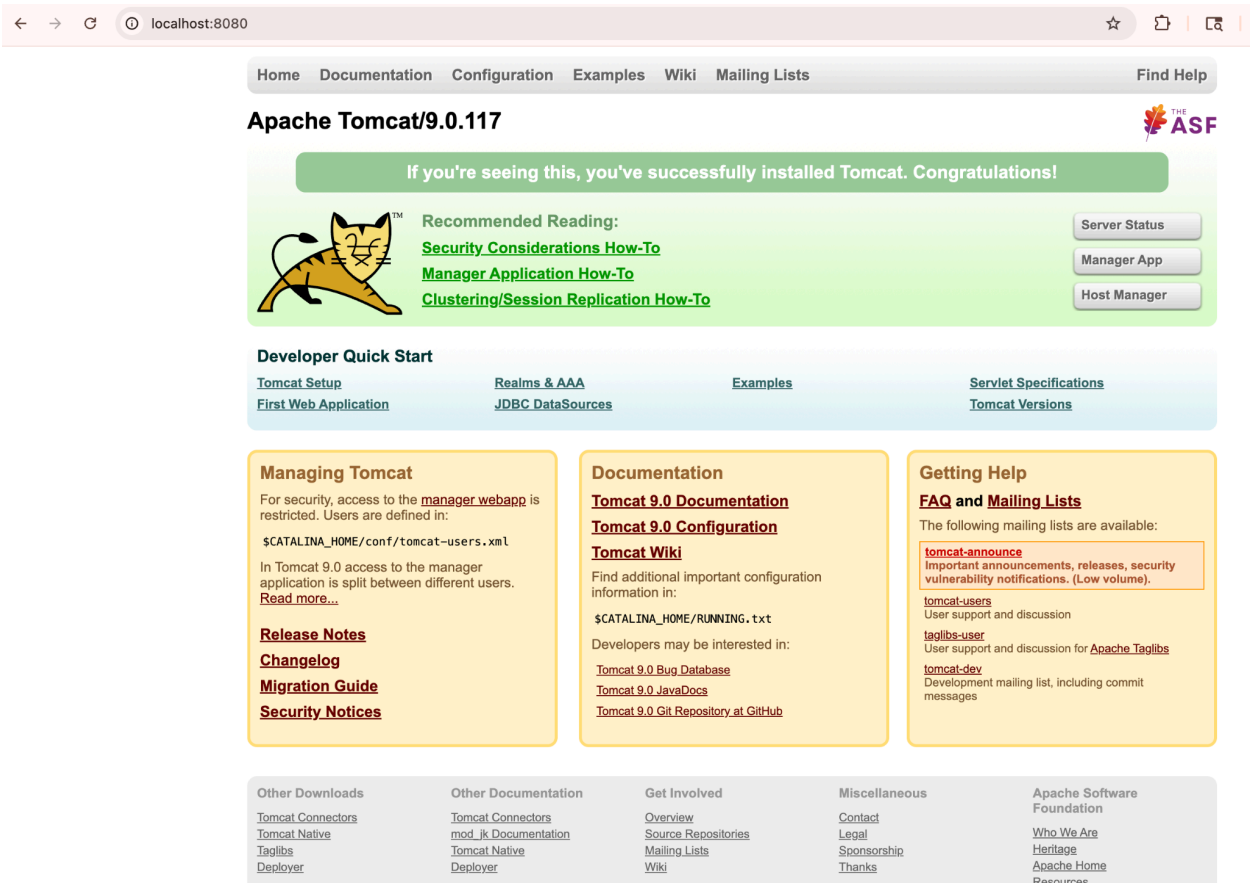
3. Verify Tomcat

Open in browser:

http://localhost:8080

Tomcat welcome page should be displayed.



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4. Check Ports Used by Java Process

```
lsof -i -P -n | grep java
```

Output:

```
dsarraf@C19536 bin % lsof -i -P -n | grep java
```

```
java      13417 dsarraf 44u  IPv6 0xbfec41bd44640209      0t0    TCP *:8080 (LISTEN)
```

```
java      13417 dsarraf   53u  IPv6 0x50f1f644013abe37      0t0    TCP 127.0.0.1:8005 (LISTEN)
```

Ports:

- **8080 → HTTP**
- **8005 → Shutdown**

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5. Remove Default Applications

```
cd ../webapps
```

```
rm -rf *
```

Restart Tomcat:

```
../bin/shutdown.sh
../bin/startup.sh
```

---

6. Deploy Jenkins WAR in Tomcat

Download:

```
wget https://get.jenkins.io/war-stable/latest/jenkins.war
```

Copy:

```
cp jenkins.war ../webapps/
```

Restart Tomcat.

---

7. Verify Jenkins

Open:

http://localhost:8080/jenkins

Jenkins setup screen should appear.

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8. Enable JMX in Tomcat

Create/edit `setenv.sh` in bin folder:

```
export CATALINA_OPTS="
-Dcom.sun.management.jmxremote
-Dcom.sun.management.jmxremote.port=1234
-Dcom.sun.management.jmxremote.rmi.port=1235
-Dcom.sun.management.jmxremote.authenticate=false
-Dcom.sun.management.jmxremote.ssl=false
"
```

Restart Tomcat.

Ports after enabling JMX:

- 8080 → HTTP
- 8005 → Shutdown
- 1234 → JMX
- 1235 → RMI

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9. Use Same Port for RMI and JMX

```
export CATALINA_OPTS="
-Dcom.sun.management.jmxremote
-Dcom.sun.management.jmxremote.port=1234
-Dcom.sun.management.jmxremote.rmi.port=1234
-Dcom.sun.management.jmxremote.authenticate=false
-Dcom.sun.management.jmxremote.ssl=false
-Djava.rmi.server.hostname=localhost
"
```

Restart Tomcat.

Ports now:

- 8080 → HTTP
- 8005 → Shutdown
- 1234 → JMX + RMI

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10. Run Tomcat with Small Heap Size

```
export CATALINA_OPTS="-Xms10m -Xmx20m"
./startup.sh
```

Expected Error:

- `java.lang.OutOfMemoryError: Java heap space`

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11. Increase Heap + Enable Parallel GC

```
export CATALINA_OPTS="-Xms1g -Xmx3g -XX:+UseParallelGC"
```

Restart Tomcat.

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12. Connect Using JConsole

```
jconsole
```

Connect to:

```
localhost:1234
```

Observe:

- Memory usage
- Threads
- CPU
- MBeans

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13. Stop Tomcat

```
./shutdown.sh
```

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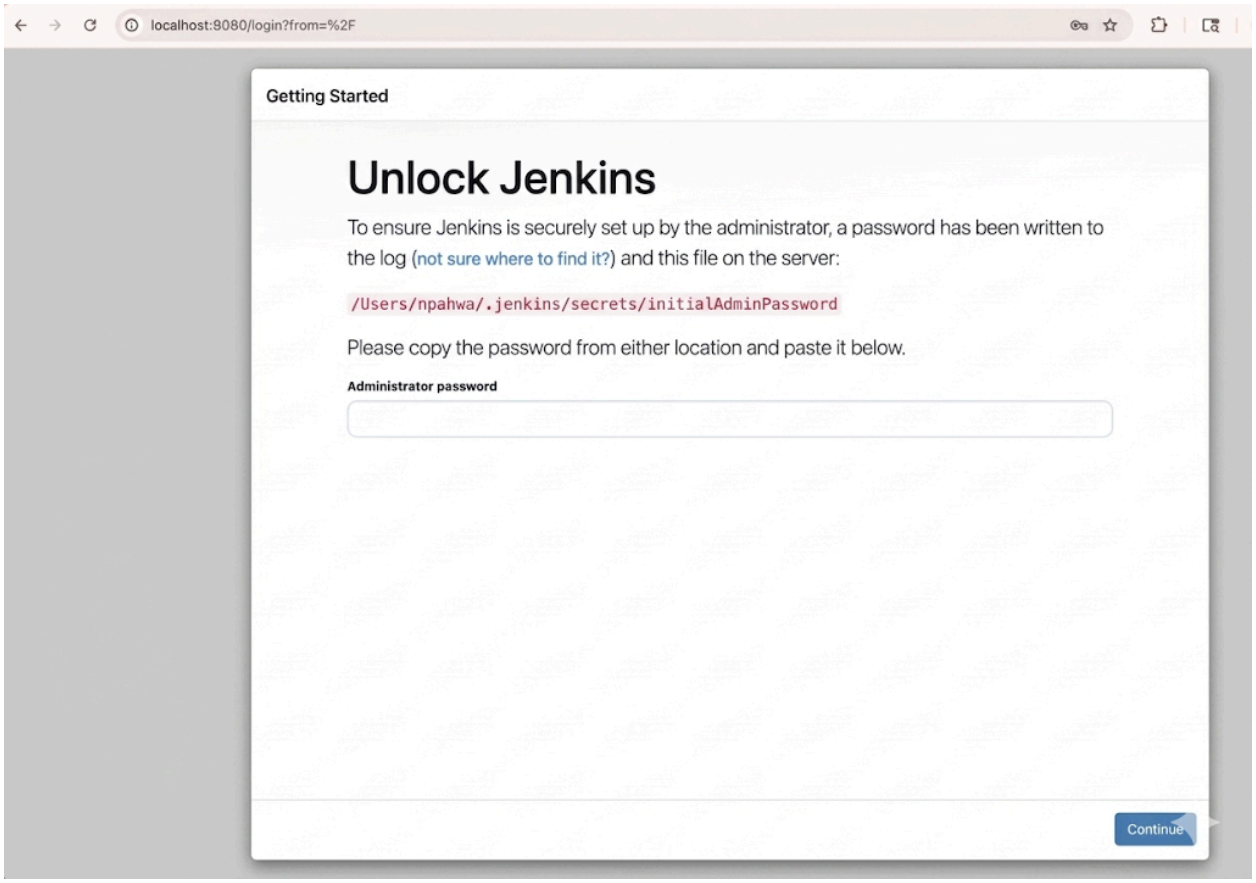
14. Run Jenkins as Standalone Application

```
java -jar jenkins.war
```

Open:
http://localhost:8080

Jenkins runs independently without Tomcat.



Key Learnings

- Java must be installed before running Tomcat
- Tomcat runs as a Java process
- Default ports: **8080, 8005**
- JMX introduces additional monitoring ports
- Small heap leads to **OutOfMemoryError**
- Jenkins can run inside Tomcat or standalone

Notes

- Use `lsof` on macOS instead of `netstat -tulnp`
- Always restart Tomcat after configuration changes

- Ensure Java environment variables are set